Which Surrogates Matter?

What Drives Success in Ensemble Transfer Attacks

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1. **What task will you study? What dataset(s) and baseline model will you start with? Identify one primary trust axis and one secondary trust axis to address.**

   Our team studies transfer-based black-box adversarial attacks on image classifiers, in which an attacker crafts adversarial examples on an ensemble of publicly available "surrogate" models and hopes they also fool an unseen target model; each of us takes a different angle on what makes some ensembles transfer better than others, and we will synthesize our findings. We will use standard image classification benchmarks (e.g., CIFAR-10, ImageNet) with a shared pool of publicly available pretrained classifiers, and a standard equal-weight ensemble attack as our common baseline. **Primary trust axis:** adversarial robustness; **secondary trust axis:** explainability.

2. **What is the proposed method you will implement, adapt, or change to address the problem?**

   My part treats the surrogates as players in a cooperative game, where the payoff of any subset is how often an ensemble attack built from it fools held-out targets; each surrogate's Shapley value is then its average contribution, and pairwise interaction values show which surrogates are redundant and which are complementary. Since the number of subsets grows exponentially, I will compute these values exactly for a small pilot pool and estimate them by sampling for larger ones. I will then test whether explanation similarity between a surrogate and the target (e.g., how alike their saliency maps are) predicts Shapley value better than simple factors like architecture type or accuracy, and use the best predictor to pick a small, effective set of surrogates without rerunning the full analysis.

3. **What evaluation metric will you use, and what stress test or sanity check will you run?**

   My main metric is the attack success rate on held-out targets, along with how well each predictor tracks the surrogates' Shapley values. To assess efficiency, I will measure how much of the full ensemble's success rate a small selected subset retains at a fraction of the compute; this builds on Shapley's efficiency axiom, under which the full ensemble's success (above the clean-image baseline) is fully distributed across the surrogates, so each Shapley value reads as that surrogate's share of the credit. As a stress test, I will evaluate on targets whose architecture type is absent from the surrogate pool; as a sanity check, I will confirm that a deliberately useless surrogate (e.g., an untrained model) receives near-zero credit, as the Shapley axioms require.

4. **What are the main data, compute, or timeline constraints? What is the biggest risk and your fallback plan?**

   The main constraint is compute, since every part of the project needs many attack runs, so we will share one attack pipeline and model pool and pilot on a small dataset before scaling up. The biggest risk is that one simple factor, such as architecture type, explains most differences in transfer; our fallback is to report how the factors compare, which still shows which surrogate properties matter most. Each part also stands on its own if another does not pan out.